# SUGUR CANE INDUSTRY DATABASE MANAGEMENT SYSTEM

SQL Database Schema

Farmers: Contains information about the farmers.

Fields: Contains information about the sugarcane fields.

Harvests: Contains information about the harvests.

```
-- Create Farmers table
CREATE TABLE Farmers (
    FarmerID INTEGER PRIMARY KEY AUTOINCREMENT,
    Name TEXT NOT NULL,
    ContactInfo TEXT,
    Address TEXT
);

-- Create Fields table
CREATE TABLE Fields (
    FieldID INTEGER PRIMARY KEY AUTOINCREMENT,
    FarmerID INTEGER,
    FieldLocation TEXT NOT NULL,
    FieldSize REAL NOT NULL,
    FOREIGN KEY (FarmerID) REFERENCES Farmers(FarmerID)
);

-- Create Harvests table
CREATE TABLE Harvests (
```

```
    HarvestID INTEGER PRIMARY KEY AUTOINCREMENT,
    FieldID INTEGER,
    HarvestDate DATE NOT NULL,
    Yield REAL NOT NULL,
    FOREIGN KEY (FieldID) REFERENCES Fields(FieldID)
);
```

**Python Script to Interact with SQLite Database**

```
import sqlite3

# Connect to SQLite database
conn = sqlite3.connect('sugarcane_industry.db')
cursor = conn.cursor()

# Create tables
cursor.execute('''
CREATE TABLE IF NOT EXISTS Farmers (
    FarmerID INTEGER PRIMARY KEY AUTOINCREMENT,
    Name TEXT NOT NULL,
    ContactInfo TEXT,
    Address TEXT
)
''')

cursor.execute('''
```

```
CREATE TABLE IF NOT EXISTS Fields (
    FieldID INTEGER PRIMARY KEY AUTOINCREMENT,
    FarmerID INTEGER,
    FieldLocation TEXT NOT NULL,
    FieldSize REAL NOT NULL,
    FOREIGN KEY (FarmerID) REFERENCES Farmers(FarmerID)
)
''')

cursor.execute('''
CREATE TABLE IF NOT EXISTS Harvests (
    HarvestID INTEGER PRIMARY KEY AUTOINCREMENT,
    FieldID INTEGER,
    HarvestDate DATE NOT NULL,
    Yield REAL NOT NULL,
    FOREIGN KEY (FieldID) REFERENCES Fields(FieldID)
)
''')

# Insert sample data
cursor.execute('''
INSERT INTO Farmers (Name, ContactInfo, Address)
VALUES ('John Doe', '555-1234', '123 Farm Road')
''')

cursor.execute('''
```

```
INSERT INTO Fields (FarmerID, FieldLocation, FieldSize)
VALUES (1, 'North Field', 10.5)
''')

cursor.execute('''
INSERT INTO Harvests (FieldID, HarvestDate, Yield)
VALUES (1, '2023-07-15', 25.3)
''')

# Commit changes and close connection
conn.commit()
conn.close()

print("Database and tables created, and sample data inserted successfully.")
```